# Gates and Circuits

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# Goals

- Identify the basic gates and describe the behavior of each
- Combine basic gates into circuits
- Describe the behavior of a gate or circuit using Boolean expressions, truth tables, and logic diagrams
- Compare and contrast a half adder and a full adder

- Gates
  - A device that performs a basic operation on electrical signals
- Circuits
  - Gates combined to perform more complicated tasks

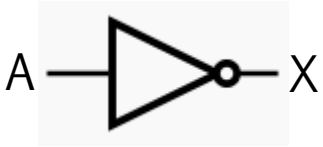
- How do we describe the behavior of gates and circuits?
  - Truth tables
    - A table showing **all possible** input value and the associated output values
  - Boolean expressions
    - Uses Boolean algebra, a mathematical notation for expressing two-valued logic
  - Logic diagrams
    - A graphical representation of a circuit; each gate has its own symbol

# Gates

- Seven types of gates
  - NOT
  - AND
  - OR
  - XOR
  - NAND
  - NOR
  - XNOR

# NOT Gate

- A NOT gate accepts one input signal (0 or 1) and returns the opposite signal as output
  - Logic Diagram Symbol




  - Boolean Expression

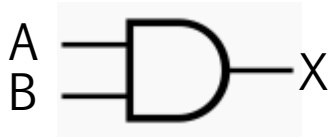
    $X = \overline{A}$ or $A'$

  - Truth Table

| A | X |
|---|---|
| 0 | 1 |
| 1 | 0 |

# AND Gate

- An AND gate accepts two input signals
- If both are 1, the output is 1; otherwise, the output is 0
  - Logic Diagram Symbol




  - Boolean Expression

$$X = A \cdot B = AB$$

  - Truth Table

| A | B | X |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

# OR Gate

- An OR gate accepts two input signals
- If both are 0, the output is 0; otherwise, the output is 1
  - Logic Diagram Symbol




  - Boolean Expression

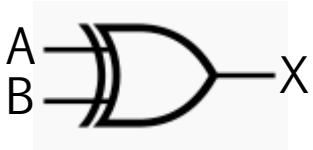
$$X = A + B$$

  - Truth Table

| A | B | X |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

# XOR (exclusive OR) Gate

- An XOR gate accepts two input signals
- If both are the same, the output is 0; otherwise, the output is 1
  - Logic Diagram Symbol




  - Boolean Expression

$$X = A \oplus B$$

  - Truth Table

| A | B | X |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

# NAND Gate

- The NAND gate accepts two input signals
- If both are 1, the output is 0; otherwise, the output is 1
  - Logic Diagram Symbol




  - Boolean Expression

$$X = \overline{A \cdot B} \quad \text{or} \quad (A \cdot B)'$$

  - Truth Table

| A | B | X |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

# NOR Gate

- The NOR gate accepts two input signals
- If both are 0, the output is 1; otherwise, the output is 0
  - Logic Diagram Symbol




  - Boolean Expression

$$X = \overline{A + B} \quad \text{or } (A+B)'$$

  - Truth Table

| A | B | X |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

# XNOR Gate

- An XNOR gate accepts two input signals
- If both are the same, the output is 1; otherwise, the output is 0
  - Logic Diagram Symbol




  - Boolean Expression

$$X = \overline{A \oplus B} \text{ or } (A \oplus B)' \text{ or } A \odot B$$

  - Truth Table

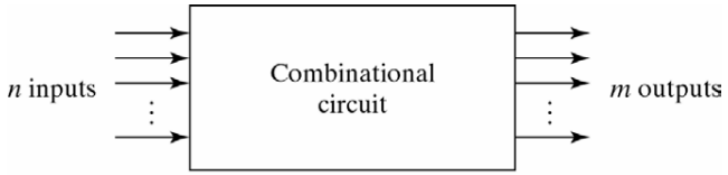
| A | B | X |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

# Review of Gate Processing

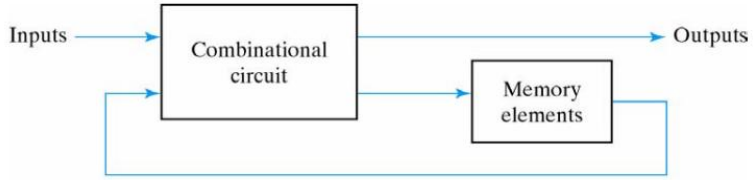
- A NOT gate inverts its single input
- An AND gate produces 1 if both input values are 1
- An OR gate produces 0 if both input values are 0
- An XOR gate produces 0 if input values are the same
- A NAND gate produces 0 if both inputs are 1
- A NOR gate produces a 1 if both inputs are 0
- An XNOR gate produces 1 if input values are the same

# Circuits

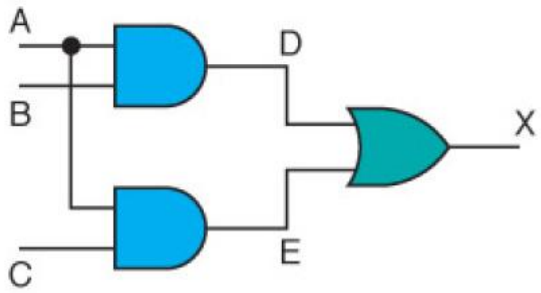
- Combinational circuit
  - Consist of logic gates whose outputs at any time are determined from the present combination of inputs
    - input variables, logic gates, and output variables

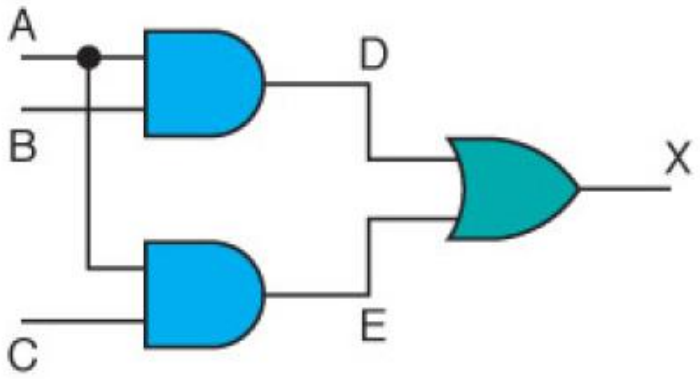



- Sequential circuit
  - Consist of memory storage elements and logic gates
    - Their outputs are a function of the inputs and the state of the storage elements



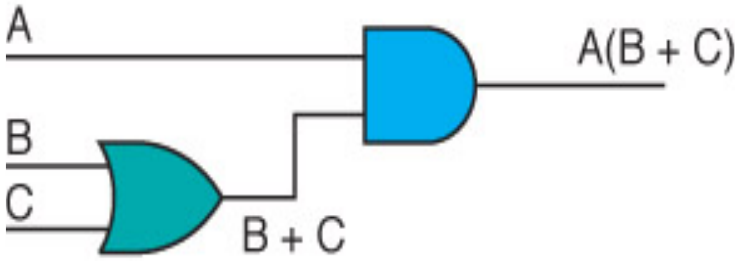
- We describe the circuit operations using
  - Boolean expressions
  - Logic diagrams
  - Truth tables
- An example of combinational circuits






| A | B | C | D | E | X |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 |

- Three inputs require eight rows to describe all possible input combinations
- This same circuit using a Boolean expression is (AB + AC)




| A | B | C | B + C | A(B + C) |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 |

| A | B | C | D | E | X |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 |

- Does this truth table look familiar?
- Compare it with previous table

- Circuit equivalence
  - Two circuits that produce the same output for identical input
- Boolean algebra allows us to apply provable mathematical principles to help design circuits
- A(B + C) = AB + AC (distributive law) so circuits must be equivalent

# Properties of Boolean Algebra

| Property | AND | OR |
|---|---|---|
| Commutative | AB = BA | A + B = B + A |
| Associative | (AB)C = A(BC) | (A + B) + C = A + (B + C) |
| Distributive | A(B + C) = (AB) + (AC) | A + (BC) = (A + B)(A + C) |
| Identity | A1 = A | A + 0 = A |
| Complement | A(A') = 0 | A + (A') = 1 |
| DeMorgan's law | (AB)' = A' OR B' | (A + B)' = A'B' |

# DeMorgan's Law

| A | B | $A \cdot B$ | $\overline{A \cdot B}$ |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 |

$$\overline{A \cdot B}$$

$$= \overline{A}\,\overline{B} + \overline{A}\,B + A\,\overline{B}$$

$$= \overline{A}\,(\overline{B} + B) + A\,\overline{B}$$

$$= \overline{A} + A\overline{B}$$

$$= \overline{A}\,(1 + \overline{B}) + A\overline{B}$$

$$= \overline{A} + \overline{A}\,\overline{B} + A\,\overline{B}$$

$$= \overline{A} + (\overline{A} + A)\overline{B}$$

$$= \overline{A} + \overline{B}$$

| A | B | A+B | $\overline{A+B}$ |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 0 |

$$\overline{A + B} = \overline{A} \cdot \overline{B}$$

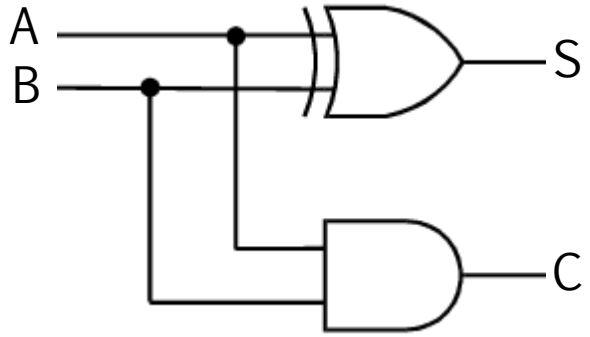
$$
\begin{aligned}
& A+B \\
&= \overline{A}\,B + A\,\overline{B} + A\,B \\
&= (\overline{A}+A)B + A\,\overline{B} \\
&= B + A\overline{B} \\
&= (1+A)B + A\overline{B} \\
&= B + AB + A\overline{B} \\
&= B + A(B+\overline{B}) \\
&= B + A \\
&= A + B
\end{aligned}
$$

# Adder

› Half Adder

| A | B | Sum (S) | Carry(C) |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |

XOR AND




$S = \overline{A}\,B + A\,\overline{B}$

$C = A\,B$

# › Full Adder

| A | B | Ci | 和(S) | 進位(Co) |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 |

$$S = \overline{A}\,\overline{B}\,Ci + \overline{A}\,B\,\overline{Ci} + A\,\overline{B}\,\overline{Ci} + A\,B\,Ci$$

$$= \overline{A}\,(\overline{B}\,Ci + B\,\overline{Ci}) + A\,(\overline{B}\,\overline{Ci} + B\,Ci)$$

$$= \overline{A}\,(B \oplus Ci) + A\,(\overline{B \oplus Ci})$$

$$= A \oplus \square$$

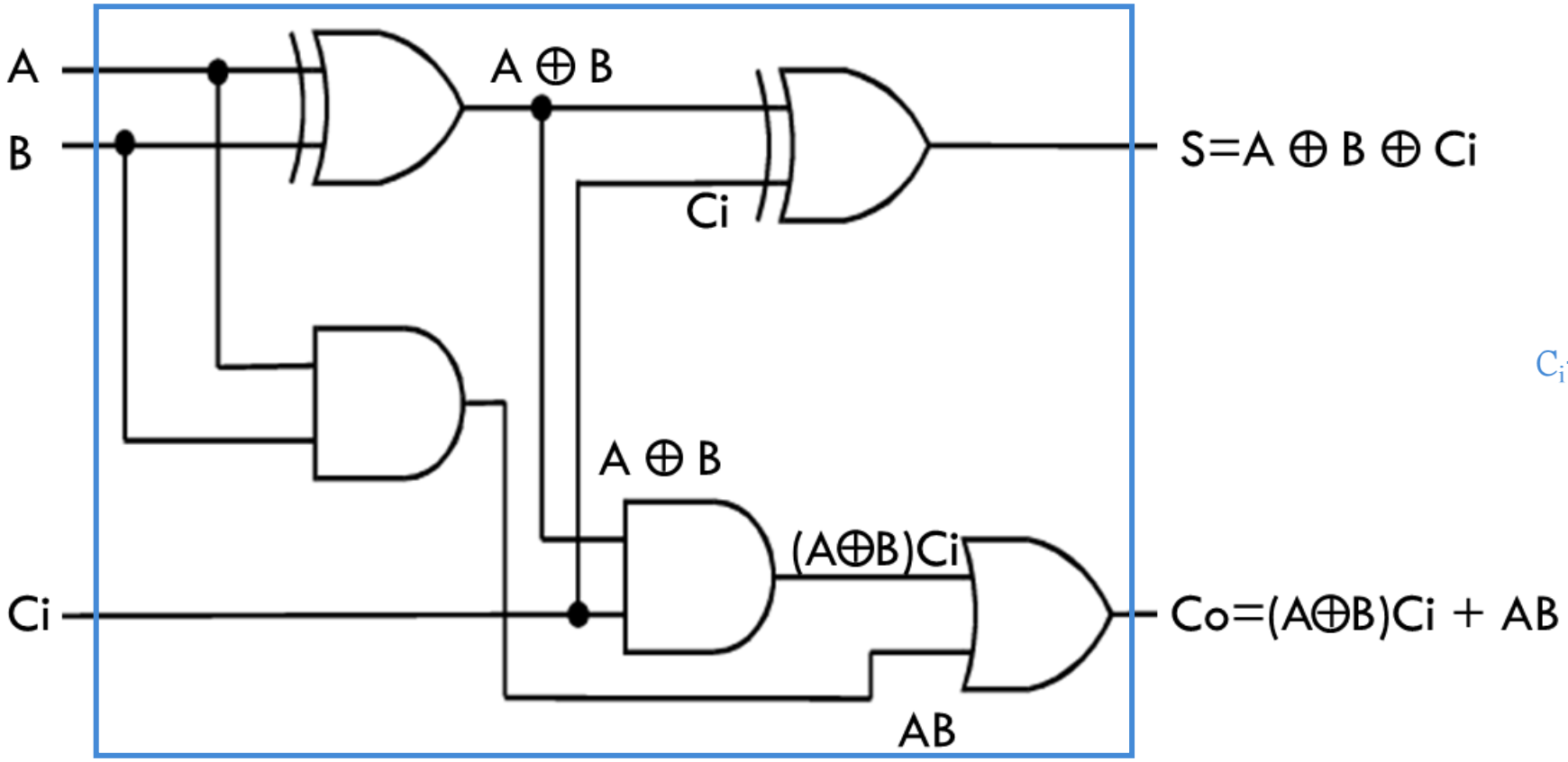
$$= A \oplus B \oplus Ci$$

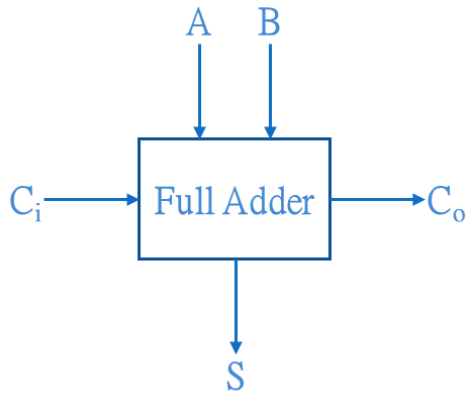
$$Co = \overline{A}\,B\,Ci + A\,\overline{B}\,Ci + A\,B\,\overline{Ci} + A\,B\,Ci$$

$$= (\overline{A}\,B + A\,\overline{B})\,Ci + A\,B\,(\overline{Ci} + Ci)$$

$$= (A \oplus B)\,Ci + A\,B$$

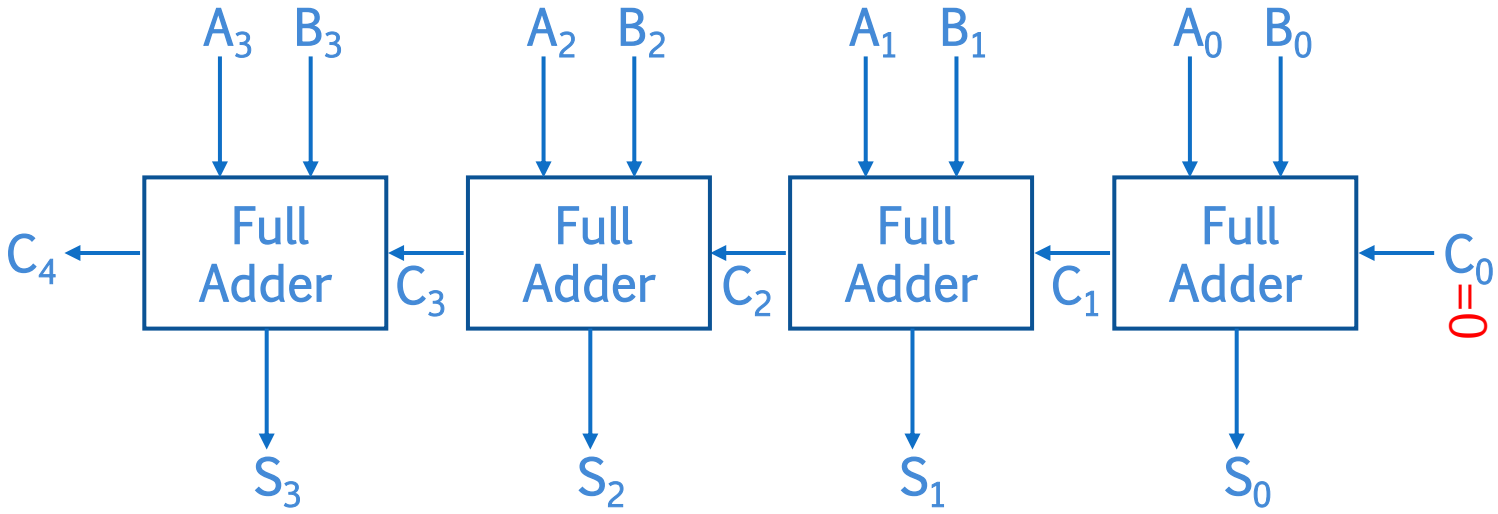
# Full Adder



› 4 bits Adder

$$
\begin{array}{r}
C_4 C_3 C_2 C_1 C_0 \\
A_3 A_2 A_1 A_0 \\
+\ B_3 B_2 B_1 B_0 \\
\hline
S_3 S_2 S_1 S_0
\end{array}
\qquad C_0 = 0
$$



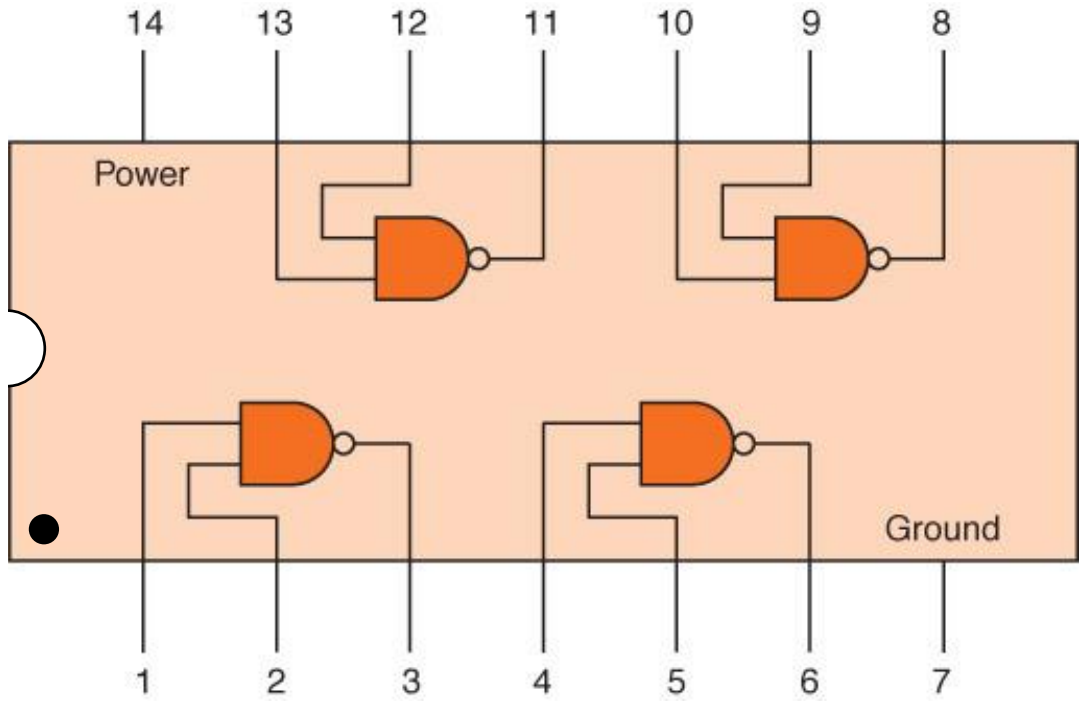
# Integrated Circuits

- Integrated circuit (IC)
  - also called a chip
- A piece of silicon on which multiple gates have been embedded
- Silicon pieces are mounted on a plastic or ceramic package with pins along the edges that can be soldered onto circuit boards or inserted into appropriate sockets
- Each pin connects to the input or output of a gate, or to power or ground





› An SSI chip contains independent NAND gates (7400)



› Integrated circuits (IC) are classified by the number of gates contained in them

| Abbreviation | Name | Number of Gates |
|---|---|---|
| SSI | Small-Scale Integration | 1 to 10 |
| MSI | Medium-Scale Integration | 10 to 100 |
| LSI | Large-Scale Integration | 100 to 100,000 |
| VLSI | Very-Large-Scale Integration | more than 100,000 |

› Q&A…